

3.3. TERM SPREAD

The decomposition reported in equation (12.7) was across the yield curve. The tax effect, however, varies with maturity. In particular, the muni curve has always been steeper than the Treasury curve, but the steepness has varied over time. This is one of the outstanding stylized facts in the muni literature and is called the *muni puzzle*.

We might expect muni yields, y_{muni} , to be related to Treasury yields, y_{tsy} , by $y_{\text{muni}} = (1 - \tau)y_{\text{tsy}}$, where τ is a relevant tax rate.¹⁴ An individual can earn 4%, say in Treasuries, but the after-tax yield she receives is $(1 - 0.396) \times 0.04 = 2.4\%$. Thus, we might expect a muni bond to deliver the same 2.4% yield to make the taxable Treasury and the tax-exempt muni investments equivalent after-tax. This relation holds at short maturities less than one year, but it fails at long maturities. Long-term muni yields are much higher than this simple tax relation predicts.¹⁵ There is a further stylized fact beyond the muni puzzle: muni yield curves have never sloped downward, while Treasuries regularly have (see chapter 9).

Various explanations have been proposed for the muni puzzle, with my sometime co-author Richard Green's (1993) among the most compelling. Green is able to construct a risk-free portfolio of taxable Treasuries so that income is transformed and treated as capital gains. Essentially, Green shows how to construct taxable bond portfolios to lessen the effect of taxes, making the muni puzzle less pronounced than it should be relative to a naïve analysis. Green modestly refers to his model as "a simple analysis" (which he puts in the title), but it requires some sophistication to follow.

Investors trade on the muni puzzle and it is called the MOB trade, for municipals-over-Treasury bonds. There are many ways to implement this strategy; a basic one is simply to buy long-term munis and sell short-term munis, hence you are exposed to the muni term spread. You fund this by taking offsetting positions in the Treasury term structure: selling long-term Treasury bonds and buying short-term Treasury notes. Then you leverage this, a lot.

Panel A of Figure 12.3 plots the twenty-year minus one-year term spread in munis (AAA) and the after-tax Treasury term spread from January 1990 to June 2011 and shows the spread for munis has been higher than the spread for Treasuries. But it has varied over time and dramatically narrowed during 2008. Panel B plots the difference between the two series. There have been pronounced increases in the muni slope relative to the after-tax Treasury slope before, as in 1992 and 1994, but nothing in the post-1990 sample is as dramatic as 2008. If the

¹⁴This is formulated by Miller (1977).

¹⁵Another way of stating the muni puzzle is that marginal tax rates implied by the ratio of the muni curve to the Treasury curve are much smaller than they should be given federal income tax rates. Estimates of implied tax rates from other muni prices, however, are high. Longstaff (2011) estimates implied tax rates around 38% using muni swaps. Ang, Bhansali, and Xing (2010) estimate marginal tax rates that exceed over 100% using muni market discount bonds.